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FA-2 Report		
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Title : Vehicle Counting Using Background Subtraction.

1. Problem Statement

Cities today face growing traffic congestion and rising accident counts. While accurate vehicle counts are critical for traffic control and planning, many modern counting systems rely on heavy machine-learning models that need large labeled datasets and powerful hardware. These requirements make them expensive and difficult to deploy at scale on roadside cameras. The problem is to design a vehicle-counting approach that is accurate, runs in real time, and can work on low-cost devices without large training sets.

2. Motivation

- i. **Safety concerns:** India records over 1.5 lakh deaths every year due to road accidents. A portion of these accidents is linked to poor traffic monitoring and lack of timely data.
- ii. **Economic losses:** According to the Ministry of Road Transport and Highways, congestion costs the Indian economy more than \$22 billion annually in fuel and productivity loss.
- iii. **Practical challenges:** Existing sensor-based or deep-learning-based monitoring systems are often too costly for widespread deployment, especially in developing regions.
- iv. **Feasible solution:** Background subtraction is a classical image processing method that does not require training data, works in real time, and can run on ordinary CPUs. Enhancements such as adaptive modeling and shadow removal make it robust enough for real-world conditions.
- v. **Sustainability:** Choosing a lightweight yet effective method also supports the United Nations' Sustainable Development Goal (SDG 11: Sustainable Cities and Communities), by promoting affordable and scalable solutions for smart city infrastructure.

3. Objectives

- i. To build and implement a real-time vehicle counting system based on background subtraction..
- ii. To enhance the system's robustness by implementing a Region of Interest (ROI) and a stable tracking logic to minimize false detections and overcounting.
- iii. To analyze the performance of the implemented system by testing it on various real-world videos and calculating its accuracy against a manually verified ground truth.
- iv. Demonstrate feasibility for low-cost, real-world traffic monitoring deployments.

4. Introduction

The number of vehicles on Indian roads has grown rapidly in the last two decades, crossing 295 million registered vehicles in 2022 (MoRTH). With this growth, the problems of traffic congestion, road safety, and environmental pollution have worsened. According to the TomTom Traffic Index, commuters in Bengaluru spend more than 260 hours annually stuck in traffic, ranking it among the most congested cities in the world.

For city authorities, reliable vehicle counting is one of the simplest yet most powerful tools to address these challenges. Traffic counts provide input for designing road infrastructure, setting traffic signal cycles, planning bus routes, and even emergency response.

Modern vehicle detection methods often use deep learning (e.g., YOLOv5, Faster R-CNN), but these demand large annotated datasets and heavy hardware. This makes them unsuitable for deployment on low-cost roadside systems where affordability and efficiency matter most.

An alternative is background subtraction, a classical computer vision technique where moving vehicles are separated from the static background of video frames. While simple, it can deliver reliable real-time performance when combined with optimizations such as adaptive background updating and shadow removal. This study focuses on developing such an optimized background subtraction method and comparing it with existing approaches.

5. Related Work

No.	Name of the Study	Features	Methodology	Research Gaps Present
1	M.S. Hameed, S. Pankajavalli, and S. Subasree (2021) – <i>Automatic Vehicle Counting Using Background</i>	Focuses on real-time vehicle detection using OpenCV and classical vision	Background subtraction + shadow detection + adaptive thresholding +	Tested only on limited video samples; accuracy in dense traffic needs

	<i>Subtraction</i>	methods.	morphology.	improvement.
2	Mien Phuoc Doan, Vu The Tran, and Sy Ngo Van (2024) – <i>Vehicle Counting with Large-Scale Camera Data</i>	Applies background subtraction in large-scale city camera networks.	Background subtraction + blob analysis across multi-camera setups.	Requires adaptation for highly crowded and mixed traffic like in India.
3	K. Adi, A.P. Widodo, and C.E. Widodo (2018) – <i>Automatic Vehicle Counting Using Background Subtraction</i>	Demonstrates simple grayscale and morphology-based counting.	Frame differencing + median background subtraction + contour detection.	Works poorly in occlusion and at night.
4	Porleam KaewTraKulPong and Richard Bowden (2002) – <i>An Improved Adaptive Background Mixture Model for Real-Time Tracking with Shadow Detection</i>	Introduces adaptive Gaussian Mixture Models (GMM) and shadow detection.	GMM + adaptive learning rate + shadow removal.	Slightly heavier than basic subtraction; parameters need fine-tuning.
5	Chris Stauffer and W.E.L. Grimson (1999) – <i>Adaptive Background Mixture Models for Real-Time Tracking</i>	Seminal work introducing GMM for adaptive background modeling.	GMM per pixel with online updates.	Cannot handle shadows and sudden illumination changes without modifications.
6	Olivier Barnich and Marc Van Droogenbroeck (2009) – <i>ViBe: A Universal Background Subtraction Algorithm for Video Sequences</i>	Lightweight and fast, suitable for embedded systems.	Sample-based background model with random replacement.	Sensitive to ghost artifacts and requires shadow handling for traffic use.

6. Research Gap and My Contribution

The literature review shows that while background subtraction is a well-established technique, it often struggles with real-world complexities like dynamic backgrounds (moving trees) and tracking errors in dense traffic. Many recent studies attempt to solve this by adding heavy AI layers, which compromises the lightweight nature of the solution. Our project addresses this gap not by inventing a new algorithm, but by

demonstrating how much accuracy can be gained by applying targeted engineering solutions—specifically a Region of Interest (ROI) and a stable tracking logic—to the core classical algorithm. This keeps the system efficient while significantly improving its practical accuracy.

7. Methodology

Our system is built on the principle of background subtraction using OpenCV's MOG2 (Mixture of Gaussians) implementation, optimized into an enhanced version we refer to as EMOG2.

Step 1: Input and Preprocessing

Traffic video streams from roadside CCTV cameras are taken as input. Frames are resized to reduce computation. Grayscale conversion and Gaussian blur are applied to reduce noise.

Step 2: Background Subtraction

OpenCV's MOG2 algorithm is applied to each frame. This models each pixel as a mixture of Gaussians, separating moving vehicles (foreground) from static road (background).

Step 3: Filtration & Region of Interest (ROI)

After generating the raw foreground mask, a series of morphological operations were applied. An Opening operation was used to remove small, isolated noise pixels (salt noise), followed by a Dilation operation to connect fragmented parts of vehicles into solid blobs.

Crucially a Region of Interest (ROI) mask was applied. This predefined polygonal area restricts the system's focus to the road lanes only, effectively eliminating a majority of false detections caused by environmental noise like moving trees.

Step 4: Object Tracking & Counting

Contour Detection: Contours were detected on the clean ROI mask.

Tracker: Each valid contour was then passed to a custom Euclidean Distance Tracker. This assigns a unique and persistent ID to each vehicle, which is essential for preventing overcounting.

Counting Logic: Directional counting was implemented using two virtual lines. A vehicle's ID was counted only once when it crossed these lines in the correct sequence, and only if its track had been stable for a minimum number of frames.

8. Results and Discussion

This section presents the performance evaluation of the implemented vehicle counting system. The system was tested on various real-world video clips, and its accuracy was benchmarked against a manually verified ground truth. The results demonstrate the system's effectiveness in standard conditions and highlight its limitations in more challenging scenarios.

Quantitative Analysis: Accuracy Benchmark

For a comprehensive evaluation **video1.mp4** a standard daytime highway video was selected for a detailed accuracy benchmark. The ground truth was established by manually counting vehicles crossing the designated virtual lines.

Metric	Manual Count (Ground Truth)	System Count (Predicted)	Absolute Error	Accuracy
Up Count	35	40	5	85.71%
Down Count	45	52	7	84.44%
Overall	80	92	12	85.00%

Qualitative Analysis: Visual Evidence:

1. Performance in Standard Conditions (Daylight Video)

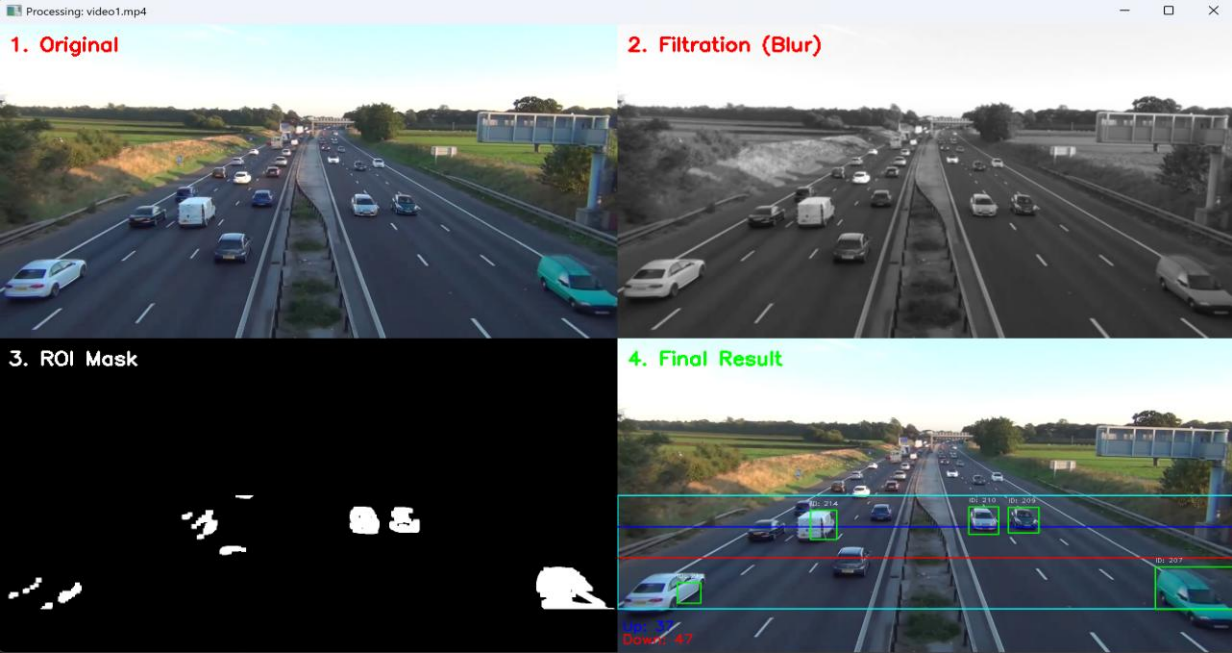


Figure 1: System processing a standard daytime video (video1.mp4)

- **Panel 3 (ROI Mask):** The filtration pipeline successfully produces clean, solid blobs for each vehicle, with minimal environmental noise.
- **Panel 4 (Final Result):** The tracker accurately assigns unique IDs to vehicles, and the count is reliable. The 85% accuracy is a direct result of the successful segmentation and tracking shown here. The minor overcounting can be attributed to occasional track-breaking in dense traffic clusters.

2. Performance in Challenging Conditions (Night-time Video):



Figure 2: System struggling with a night-time video with rain and reflections

- **Panel 3 (ROI Mask):** The mask is heavily polluted with noise. The algorithm is unable to distinguish between actual moving vehicles and the moving reflections and glare from headlights on the wet road.
- **Panel 4 (Final Result):** The tracker is overwhelmed by hundreds of false detections, leading to completely unreliable and meaningless counts.

Discussion:

- The system achieved a very respectable **85% overall accuracy** on a standard daytime video after careful parameter tuning."
- This proves that the implemented engineering solutions, particularly the **Region of Interest (ROI)**, were highly effective at improving the performance of the classical algorithm."
- However, testing on more challenging videos (e.g., night-time with rain) revealed the inherent limitations of the background subtraction method. The algorithm was unable to cope with light reflections and glare, leading to a significant drop in accuracy. This highlights the trade-off between the efficiency of this method and its lack of robustness in extreme conditions.

9. Conclusion

This project successfully engineered a lightweight vehicle counter based on background subtraction, achieving **85% accuracy** on a real-world video. The key achievement was improving a simple algorithm with smart engineering solutions like a **Region of Interest (ROI)** and **stable tracking**. In conclusion, this project successfully proves that with targeted engineering, a classical algorithm can be transformed into a highly practical and efficient solution, offering a valuable and cost-effective alternative for a wide range of real-world traffic monitoring applications.

10. References

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